

A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill is faintly visible in the background.

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DRUG WITHDRAWAL SYNDROMES

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

AA Alcoholics Anonymous

AWS Alcohol Withdrawal Syndrome

BPM Beats Per Minute

BZD Benzodiazepine

CBT Cognitive Behavioural Therapy

CIWA-Ar Clinical Institute Withdrawal Assessment for Alcohol Withdrawal

CIWA-B Clinical Institute Withdrawal Assessment for Benzodiazepines

COWS Clinical Opiate Withdrawal Scale

DSM Diagnostic and Statistical Manual of Mental Disorders

DT Delirium Tremens

GERD Gastroesophageal Reflux Disease

HR Heart Rate

N Nausea

OTC Over-The-Counter

PRN As Needed

QTc Corrected QT Interval

SAS Sedation-Agitation Scale

V Vomiting

OBJECTIVES

- 1 Describe characteristics of drug withdrawal syndromes
- 2 Identify signs and symptoms of drug withdrawal syndromes
- 3 Understand the importance of support and counselling
- 4 Determine appropriate therapy

DEFINITION

Physical Dependence (e.g. nausea, vomiting, diarrhea, seizures, hallucinations)

- Physiological adaptation to a substance in the central nervous system due to long-term exposure to a substance to obtain a particular effect

Tolerance

- Need for increasing amounts of drug to achieve desired effects or markedly decreased effect of drug with continuous use of same dose

Drug Withdrawal Syndromes

- Occurs due to reduction or cessation of a drug after a period of heavy and prolonged use
- May cause psychological and physical symptoms

Drug/Substance Dependence

- Psychological and physical adaptation that is characterized by maladaptive patterns of use, resulting in significant impairment or distress

ALCOHOL WITHDRAWAL SYNDROME (AWS)

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CLINICAL PRESENTATION

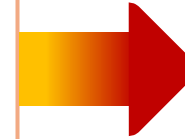
Complications:

- Thiamine (Vitamin B1) deficiency
- Low electrolytes
 - Potassium
 - Magnesium
 - Phosphorous
- Liver disease
- Wernicke encephalopathy
 - Mental confusion
 - Ataxia
 - Ophthalmoplegia

Within **hours to days** after cessation/reduction:

Mild

- Insomnia
- Tremors
- Anxiety
- Diaphoresis
- Palpitations
- Nausea
- Psychomotor agitation



Moderate-Severe

- Withdrawal seizures
- Hallucinations
- Delirium Tremens (DT)
 - Mortality rate ~ 5%

DIAGNOSIS

According to the **DSM-5**, the following criteria is required for **AWS diagnosis**:

- Discontinuation (or reduction) of heavy and prolonged alcohol use
- Clinically significant distress or functional impairment (e.g. social, occupational)
- No other cause of signs and symptoms
- At least **2 of the following symptoms**, hours to days after cessation of alcohol use:
 - Autonomic hyperactivity (diaphoresis, pulse >100bpm)
 - Increased hand tremor
 - Insomnia
 - Nausea/vomiting
 - Transient hallucinations
 - Psychomotor agitation
 - Anxiety
 - Grand mal seizures

DIAGNOSIS (CONTINUED)

- Detailed history and physical examination
 - Assess use of alcohol/other substances
 - Assess functional impairment
 - Rule out differential diagnoses and identify co-morbidities
- Lab tests, imaging, toxicology tests
- Clinical Institute Withdrawal Assessment for Alcohol (CIWA-Ar)
 - Assess severity of alcohol withdrawal
- Sedation-Agitation Scale (SAS)
 - Assess agitation

GOALS OF THERAPY

- ✓ Assess and manage drug withdrawal symptoms
- ✓ Prevent mortality from drug withdrawal
- ✓ Change pattern of alcohol use based on patient goals
- ✓ Prevent pharmacotherapy-related adverse effects
- ✓ Prevent relapse

NON-PHARMACOLOGICAL MEASURES

Prevention

- Motivational interviewing and stages of change
- Alcoholics Anonymous (AA)

Treatment

- Non-judgmental approach
- Psychosocial therapy (e.g. CBT and motivational enhancement therapy)

PHARMACOLOGICAL THERAPY: SUPPORTIVE CARE

IV fluids in cases of volume deficit

- Excessive vomiting, sweating or diarrhea

Replenish electrolytes

- Commonly, patients have low magnesium, potassium, and phosphorous

Nutrition supplementation: routinely given to all patients with AWS

- **Thiamine** 100-200 mg IV, IM, PO daily
 - Prevents Wernicke's encephalopathy
- **Glucose**
 - Given concurrently to prevent Wernicke's encephalopathy
- **Multivitamins**
 - Corrects deficiencies
- **Folate**

PHARMACOLOGICAL THERAPY: BENZODIAZEPINES

- **First-line therapy** for alcohol withdrawal in patients at **high risk** of severe complications of alcohol withdrawal
 - Effective therapy to prevent seizures, hallucinations, agitation, tremors, autonomic hyperactivity
 - MOA: Modulators of GABA receptors
 - Side effects: ataxia, dizziness, lightheadedness, weakness
 - Treatment should be short-term and limited to the acute phase

Dosing:

- Medication is **administered based on CIWA-Ar score** (reassessment of score at repeated intervals) (symptom-triggered dosing)
 - Counsel patient on risk of respiratory depression if relapse occurs soon after benzodiazepine treatment
- **Alternative fixed-dosing** schedule
 - Set intervals and dose, even if asymptomatic
 - Taper dose at certain intervals
 - Infrequent reassessments, useful in high-risk patients

PHARMACOLOGICAL THERAPY (Continued)

Intermediate-acting benzodiazepines

- More frequent dosing
- More suitable for the **elderly and those with liver dysfunction**
 - Lorazepam
 - Oxazepam

Long-acting benzodiazepines

- Preferred for the general adult population
- **Less risk of rebound effects or withdrawal seizures**
 - Chlordiazepoxide
 - Diazepam

****** If response is not adequate (resistance case), can add-on phenobarbital or increase the dose of the benzodiazepine

- If the patient is at **low risk** of severe alcohol withdrawal complications, they can be managed in an outpatient setting with **gabapentin, carbamazepine and/or adjunctive clonidine**

CASE #1

KY is a 56-year-old male who presents to the emergency department with complaints of severe vomiting, diarrhea, anxiety, tremors, and sweating. He denies hallucinating and he has not experienced any seizures. These symptoms began approximately 6 hours after his last drink. His wife is with him and explains to the on-call staff that he has a history of severe alcohol use, drinking approximately 4-5 glasses of alcohol daily as he has done for the past 3 years. She further explains that he made the difficult decision to quit cold-turkey after his cousin was severely injured due to an incident that involved driving under the influence. His past medical history is significant for hypertension and gastroesophageal reflux disease (GERD). His current medications include amlodipine 5 mg PO daily and pantoprazole 40 mg PO daily. He has no known allergies to any medications. KY's wife also tells the medical team that he has no history of recreational drug use and that he has never smoked. His vitals are checked and reveal a heart rate of 112 beats per minute (bpm), a blood pressure of 154/92 mmHg, a temperature reading of 37.6°C, a respiratory rate of 20 breaths per minute, and an arterial blood O₂ saturation of 98%. Laboratory test results reveal a serum aspartate aminotransferase (AST) level of 35 units/L (reference: 10-40 units/L), a serum alanine aminotransferase (ALT) level of 28 units/L (reference: 10-40 units/L), a serum 25-hydroxyvitamin D level of 102 nmol/L (reference: 75-150 nmol/L), and a serum vitamin B12 level of 331 pmol/L (reference: 147.5 to 590 pmol/L).

CASE #1 - QUESTIONS

1. Which of the following pharmacologic agents is the most appropriate to administer to KY?
 - a) Lorazepam
 - b) Diazepam
 - c) Temazepam
 - d) Triazolam
2. Which of the following supplements is routinely given to patients experiencing alcohol withdrawal syndrome?
 - a) Vitamin D
 - b) Vitamin C
 - c) Vitamin B7
 - d) Vitamin B1
3. Which of the following statements correctly identifies the reason glucose is given to patients experiencing alcohol withdrawal?
 - a) To prevent delirium tremens
 - b) To prevent seizures
 - c) To prevent Wernicke's encephalopathy
 - d) To counteract high insulin levels
4. Which of the following electrolyte abnormalities is LEAST important to check in KY?
 - a) Sodium
 - b) Potassium
 - c) Magnesium
 - d) Phosphorous

OPIOID WITHDRAWAL

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CLINICAL PRESENTATION

Short-acting opioids (e.g. heroin)

- Withdrawal begins 8-24 hours after last dose

Long-acting opioids (e.g. methadone)

- Withdrawal begins 12-48 hours after last dose

The severity of withdrawal symptoms and duration of withdrawal symptoms (usually lasting for 4-20 days) is dependent on opioid half-life and the duration/amount of opioid use (tolerance)

- Autonomic Hyperactivity
- Dysphoria
- Anxiety
- Insomnia
- Yawning
- Restlessness
- Rhinorrhea
- Lacrimation
- Myalgia
- Nausea and vomiting
- Diaphoresis
- Abdominal cramps
- Diarrhea
- Piloerection
- Kicking movements and spasms ("kicking the habit")



Examples of opioids

- Methadone
- Fentanyl
- Hydromorphone
- Morphine
- Oxycodone
- Heroin

DIAGNOSIS

- Detailed history and physical examination
- Lab tests, imaging, toxicology tests
- Clinical presentation
- Clinical Opiate Withdrawal Scale (COWS)
 - Assess severity of withdrawal

According to the **DSM-5**, the following criteria is required

- Symptoms develop secondary to stopping or reducing prolonged opioid use
 - OR symptoms develop after opioid antagonist is given after prolonged opioid use
- Clinically significant distress or functional impairment
- No other cause of signs and symptoms
- At least **3 of the following symptoms**: dysphoria, nausea, vomiting, myalgia, fever, lacrimation, rhinorrhea, insomnia, diarrhea, yawning, pupil dilation, piloerection, diaphoresis

GOALS OF THERAPY

- ✓ Prevent withdrawal complications
- ✓ Relieve withdrawal symptoms
- ✓ Prevent relapse

NON- PHARMACOLOGICAL MEASURES

- Psychosocial treatments (e.g. CBT and motivational enhancement therapy)
- Counselling with nonjudgmental approach

DRUG THERAPY: OPIOID PARTIAL AGONIST

BUPRENORPHINE/NALOXONE (SUBOXONE®)

First-line therapy for detoxification and maintenance

- Dosage form: Sublingual tablet (takes approximately 5 min to dissolve)
- Partial agonist at mu opioid receptor
- Usually dosed every 1-2 days
- Naloxone added to prevent abuse (injection/intranasal misuse); poor oral bioavailability
- Considered **safer than methadone** (lower risk of respiratory depression and drug interactions)
- Started in patients who are in **moderate opioid withdrawal**
- Side effects
 - Sweating, headache, abdominal pain, N/V/constipation, insomnia, respiratory depression
 - **Can precipitate opioid withdrawal symptoms**

DRUG THERAPY: METHADONE (OPIOID AGONIST)

Alternative for detoxification and maintenance therapy

- Dosage form: Liquid commonly mixed with orange juice
- Long-acting: 24-36 hours; dosed daily
- No ceiling dose
- Higher maintenance doses (60-100 mg) associated with better outcomes
- **Methadone Maintenance Treatment Program**
 - Initial doses must be taken under direct supervision of pharmacists
 - May receive 'carries' to bring home, once clinically stable
- Side effects
 - Sedation, hypotension, weight gain, N/V/constipation, sweating, QTc prolongation, respiratory depression

DRUG THERAPY: A2-ADRENERGIC AGONIST: CLONIDINE

Non-opioid alternative for withdrawal symptoms

- Dosage form: Oral tablet
- ↓ sympathetic activity
- Use when methadone and buprenorphine are contraindicated/unavailable
- Blunts autonomic withdrawal symptoms (e.g. sweating, abdominal cramps, chills, anxiety, insomnia, and tremor); reduces BP and HR
- Monitor blood pressure and heart rate (prior to administration)
- Taper slowly after prolonged use due to risk of rebound hypertension
- Side effects
 - Sedation, dry mouth, orthostatic hypotension, dizziness

DRUG THERAPY: SUPPORTIVE THERAPIES

Non-opioid alternatives for withdrawal symptoms

Nausea

- Dimenhydrinate 50-100 mg Q6H PRN, Gravol® Ginger
- Haloperidol 0.5-1 mg PO Q8-12H
- Prochlorperazine 5-10 mg PO Q6-8H
- Nabilone 0.25-0.5 mg PO HS up to 0.5-1 mg PO TID

Diarrhea

- Loperamide

Insomnia

- Sleep hygiene measures
- Short-term: trazodone 25-100 mg PO HS, amitriptyline 10 mg PO HS

Myalgia

- NSAIDs (e.g., naproxen, ibuprofen)
- Acetaminophen 650-1000 mg PO Q6H as needed

CASE #2

SM is a 26-year-old female who presents to the clinic with symptoms of nausea, vomiting, diarrhea, myalgia, and restlessness. She tells the walk-in doctor that she has a constant urge to kick her leg throughout the day. These symptoms began approximately 20 hours after her last dose of an opioid analgesic medication. On further questioning, she tells the physician that she has been taking oxycodone 10 mg/acetaminophen 325 mg PO Q4-6H for the past year following a motor vehicle collision injury. After watching several documentaries on the risk of an opioid overdose and the opioid epidemic, she decided to try and stop taking the medication by herself without consulting her family physician first. Her past medical history is otherwise significant for generalized anxiety disorder for which she takes escitalopram 10 mg PO daily (initiated 2 years ago). She takes no other over-the-counter (OTC) products or supplements. She used to drink alcohol but has been abstinent for approximately 2 years, and she has no history of recreational drug use. Her BMI is 23 kg/m².

Her vitals are checked and reveal a heart rate (HR) of 121 beats per minute (bpm), a respiratory rate of 23 breaths per minute, a temperature reading of 37.1°C, and a sitting-blood pressure of 130/82 mmHg. A physical examination reveals dilated pupils with lacrimation and signs of mild agitation.

CASE #2 - QUESTIONS

1. What is the most appropriate explanation for SM's symptoms?
 - a) Opioid withdrawal
 - b) Serotonin syndrome
 - c) Opioid overdose
 - d) Alcohol withdrawal
2. Which of the following agents is LEAST appropriate to initiate for SM?
 - a) Methadone
 - b) Buprenorphine/naloxone
 - c) Clonidine
 - d) Acetaminophen
3. Which of the following statements correctly identifies why buprenorphine is co-formulated with naloxone?
 - a) To increase the analgesic potency
 - b) To reduce the risk of respiratory depression
 - c) Deter intravenous misuse
 - d) To enhance the absorption following sublingual administration
4. Which of the following is the most appropriate next step in SM's long-term management once her withdrawal symptoms have stabilized?
 - a) Discontinue all medications and refer SM for abstinence-based counselling
 - b) Recommend OTC medications for breakthrough cravings PRN
 - c) Long-term opioid agonist therapy
 - d) Initiate oral naltrexone/bupropion

BENZODIAZEPINE WITHDRAWAL

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BENZODIAZEPINES

- Benzodiazepines (BZD) are CNS depressants used to treat anxiety, insomnia, withdrawal syndromes, and seizures
- Common sedative-hypnotic agent that can result in withdrawal symptoms upon sudden discontinuation

Short-acting benzodiazepines* include

- Midazolam
- Triazolam

Intermediate-acting benzodiazepines include

- Temazepam
- Lorazepam
- Nitrazepam
- Oxazepam

Long-acting benzodiazepines** include

- Diazepam
- Chlordiazepoxide
- Flurazepam
- Clonazepam

* Short-acting benzodiazepines: withdrawal typically begins 1-2 days after the last dose, and continues for 2-4 weeks or longer

** Long-acting benzodiazepines: withdrawal typically begins 2-7 days after the last dose, and continues for 2-8 weeks or longer

→ Generally, milder symptoms with long-acting BZD than with short-acting BZD

CLINICAL PRESENTATION

Sedative/Hypnotics Withdrawal Symptoms:

- Insomnia
- Anxiety
- Hand tremors
- GI discomfort
- Psychomotor agitation
- Nausea / Vomiting
- Muscle tension and aches
- Photophobia (discomfort to bright lights)
- Phonophobia (discomfort to environmental sounds)
- Confusion
- Depersonalization (feeling detached from mind or body)
- Delirium
- Seizures
- Poor concentration and memory

DIAGNOSIS

- Medical history, clinical presentation and physical examination
- Clinical Institute Withdrawal Assessment for Benzodiazepines (CIWA- B)
 - Assessment tool to help make clinical decisions
- According to the **DSM-5**, the following criteria is required for benzodiazepine withdrawal syndrome diagnosis:
 - Syndrome develops secondary to stopping or reducing prolonged benzodiazepine use
 - Clinically significant distress or impairment of functioning
 - No other cause of disorder
 - At least **2 of the following symptoms**:
 - Autonomic hyperactivity
 - Tremor of hands
 - Insomnia
 - Transient hallucinations
 - Nausea/vomiting
 - Psychomotor agitation
 - Grand mal seizures
 - Anxiety

GOALS OF THERAPY

- ✓ Prevent withdrawal complications and mortality
- ✓ Relieve withdrawal symptoms
- ✓ Prevent relapse

THERAPEUTIC APPROACH

- Structured discontinuation program with monitoring required
- In acute BZD withdrawal:
 - Titrate to effect with long-acting benzodiazepine
 - Diazepam IV or PO depending on severity of withdrawal
- Once stabilized, taper BZD over few months
 - At least one week between dose reductions.
 - Rate of taper depends on the initial dose of BZD (high versus lower doses)
- For patients on prolonged BZD therapy, taper over 6 – 12 weeks
 - Effective in patients with drug dependence and not abusing other substances
 - Patients not being actively treated for anxiety disorders
- For patients on short-acting BZD, multiple agents, or at high risk of misuse, substitute with long-acting BZD then taper
 - Individualize based on patient factors (e.g. avoid long-acting BZDs in ≥ 65 due to accumulation risk)

CASE #3

OP is a 62-year-old male who presents to the emergency department with complaints of severe anxiety, muscle tension, hand tremors, confusion, and insomnia. He was previously taking triazolam 0.25 mg PO HS for sleep-onset insomnia, however, due to pressures from his family he decided to quit “cold-turkey”. His symptoms began one the first night that he did not use the medication for his insomnia. His past medical history is significant for dyslipidemia, benign prostatic hyperplasia, and intermittent restless legs syndrome that occurs at night. His current medications include rosuvastatin 10 mg PO daily, tamsulosin 0.4 mg PO daily and a multivitamin supplement PO daily. He has no known allergies to any medications, and he denies any use of alcohol or tobacco.

His vitals show a heart rate (HR) of 110 beats per minute (bpm), a blood pressure reading of 145/91 mmHg, and a respiratory rate of 19 breaths per minute.

CASE #3 - QUESTIONS

1. What is the most important serious complication associated with benzodiazepine withdrawal?
 - a) Insomnia
 - b) Seizures
 - c) Delirium tremens
 - d) Bradycardia
2. Which of the following agents is the most appropriate to initiate in OP?
 - a) Administer flumazenil IV
 - b) Initiate a tapering schedule using diazepam
 - c) Restart triazolam at his previously prescribed dose
 - d) Initiate propranolol to reduce tremors
3. Following the appropriate management step, OP wants to know how long he will be required to take the medication for. What is the most appropriate duration to tell OP?
 - a) Up to 24 hours
 - b) 2-7 days
 - c) Several months
 - d) 2-3 years
4. Several weeks later, OP reports that he is experiencing issues with his insomnia and intermittent restless legs syndrome. What is the most appropriate therapeutic agent to recommend?
 - a) Levodopa-carbidopa
 - b) Midazolam
 - c) Gabapentin
 - d) Temazepam

CNS STIMULANT WITHDRAWAL

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CNS STIMULANTS WITHDRAWAL

MOA: Stimulates CNS to produce dopamine and norepinephrine

CNS stimulants include

- Cocaine/Crack
- Amphetamines
- Methamphetamine ("Speed", "Meth")

* Symptoms begin within 24 hours of last use of stimulants and last for 3-5 days

CLINICAL PRESENTATION

- Intense cravings
- Agitation
- Insomnia
- Fatigue
- Increased appetite
- Depression
- Altered mental status
- Unpleasant, vivid dreams

DIAGNOSIS

- Medical history
- Clinical presentation
- Physical examination

According to the **DSM-5**, the following criteria is required for stimulant withdrawal syndrome diagnosis:

- Syndrome develops secondary to stopping or reducing prolonged stimulant use
- Clinically significant distress or impairment of functioning
- No other cause of disorder
- Dysphoric mood plus at least **2 of the following symptoms**:
 - Fatigue
 - Increased appetite
 - Vivid or unpleasant dreams
 - Insomnia or hypersomnia
 - Psychomotor retardation or agitation

GOALS OF THERAPY

- ✓ Cessation of drug use
- ✓ Prevent withdrawal complications
- ✓ Prevent relapse

THERAPEUTIC APPROACH

- There is **currently no approved medication** for the management of cocaine or methamphetamine dependence
 - Dopamine agonists and antidepressants produce conflicting results
 - **Methylphenidate may be used as maintenance therapy** under the physician's discretion (due to significant potential for abuse)
- Symptomatic medication and supportive care
- Behavioral management
- Psychological therapies (Cognitive-behavioural therapies)

CASE #4

LZ is a 42-year-old female who was previously taking methamphetamine chronically for the past 5 years. After running into financial constraints, she was unable to obtain her usual supply of methamphetamine and is currently experiencing symptoms of an increased appetite, fatigue, depression, and dreams that feel intensely real. She denies having suicidal ideation. Her last dose of methamphetamine was 4 days ago. She explains that the symptoms started approximately 36 hours after her last dose and that they have gradually worsened since. Her past medical history is insignificant, and she takes no other medications other than occasionally using cannabis 1-2 times per week.

Her current heart rate (HR) is recorded as 72 beats per minute, her blood pressure is 118/74 mmHg, and her temperature is 36.9°C.

CASE #4 - QUESTIONS

1. Which of the following symptoms is LEAST likely to occur following CNS stimulant withdrawal?
 - a) Psychomotor retardation
 - b) Insomnia
 - c) Euphoria
 - d) Altered mental status
2. Which of the following counselling points is important to include if the physician decides to initiate methylphenidate (Concerta®) in LZ?
 - a) Take the medication at bedtime to address your symptoms of vivid dreams
 - b) Report new or worsening symptoms of mood changes, agitation or suicidal thoughts to your doctor
 - c) If a dose is missed, take two pills at the same time if the next dose is due
 - d) Crush or split the tablet if you have trouble swallowing
3. How long are LZ's symptoms expected to last for since they began?
 - a) 5 days
 - b) 14 days
 - c) Several months
 - d) 1-2 years
4. Which of the following non-pharmacologic interventions is the LEAST appropriate to recommend for LZ?
 - a) Group drug counselling
 - b) Motivational interviewing
 - c) High-intensity physical activity
 - d) Cognitive-behavioural therapy (CBT)

SUMMARY OF TREATMENTS

Alcohol withdrawal

- Supportive care (IV fluids, electrolytes, thiamine, glucose)
- Benzodiazepine

Opioid withdrawal

- Buprenorphine
- Methadone

Benzodiazepine withdrawal

- Titrate with long-acting benzodiazepine
- Taper off over a few months

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CHANGE LOG

June/July 2025

- Grammar and formatting edits throughout
- Slide 3: Updated legend
- Slide 29: Added clinical presentation information
- Slide 31: Combined 2 slides into one
- Slides 15, 16, 26, 27, 34, 35, 41, 42: Case study and questions added
- Minor formatting and wording changes made
- Slide 2: Updated NAPRA competencies

November 2025

- Minor formatting changes made

March 2026

- Minor formatting and wording changes made
- Slide 2: Updated NAPRA competencies
- Slide 12: Added folate
- Slides 13 & 14: Updated information and added treatment options for patients at low risk of severe alcohol withdrawal complications
- Slide 44: Updated references